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Part 1: Design and implementation of database system

Requirement a:

Consulting with the client

- V: Hello, Ms Harbour. Please tell me what do you want from the system you want us to design.
- H: Greetings. I desire a system that emulates the feeling of buying from a market, like the old days.
- V: Can you be more specific? I don't think I quite follow.
- H: So, they should be able to simply visit the website, choose the item from a list of items from different manufacturers and delivery services, and simply be able to buy them. And the system will have to store the data of all parties like the requirement I gave you before.
- V: So, A simple system that requires no log in, integrating the manufacturer and delivery service name inside of the item properties, with delivery, customer and supplier having addresses.
- H: Yes, and moreover, the client can buy multiple items of the same type and multiple types, all in one order.
- V: I see. Is that all?
- H: I'm quite sure it is. Thank you.
- V: Thank you as well. I will get to work now.

Requirement Identification:

1. Functional:
    - A system able to process and store the transaction of items and batch item purchase, then send the information of its client, including phone, name and address, to the Delivery service
    - Allows the customer to pick the supplier through Supplier-based item sorting (there can be multiple items of the same supplier or one item of different suppliers). The system would allow the Supplier to update the stock of items available.
    - Allows the system to store operable address and shipping information of the parties involved in simple and consistent format while seperating different audiences.
2. Non functional:
    - Fast, simple, easy to manage.
    - Allows partial user anonymity, making the buying experience simpler and less time consuming.

Identification

- Firstly, the three human entities are created first: Customer, Supplier and Delivery, each containing Name, Phone and Address properties.
- Afterwards, a Item Entity is created, containing Name, Description, Count (remaining), Cost and Manufacturer properties. The Item receives a Foreign Key of SupplierID in order to differentiate products that has multiple suppliers, and since a Supplier can have many different items, the relationship between Supplier and Item is One to Many.
- Finally, the Sale Entity is created in order to Manage sales, with links to the Item, Customer and Delivery entities in order to manage the different parts of the Sale. One Delivery service can have multiple Sales, but one Sale can not have multiple Delivery Service, same with Item and Customer, so the relationship between the three of them and Sale is One to Many. Moreover, the Sale entity has a Item Count property, indicating how many items of the kind has been brought.

Requirement b:

With the Identification of Entities done in the first requirement, the following is the initial ERD:

Requirement c:

- 0NF to 1NF: Atomize properties:
    - Example 1: Name property of Customer Table as one unit -> Seperated into 3 different properties: First Name, Second Name, Last Name,

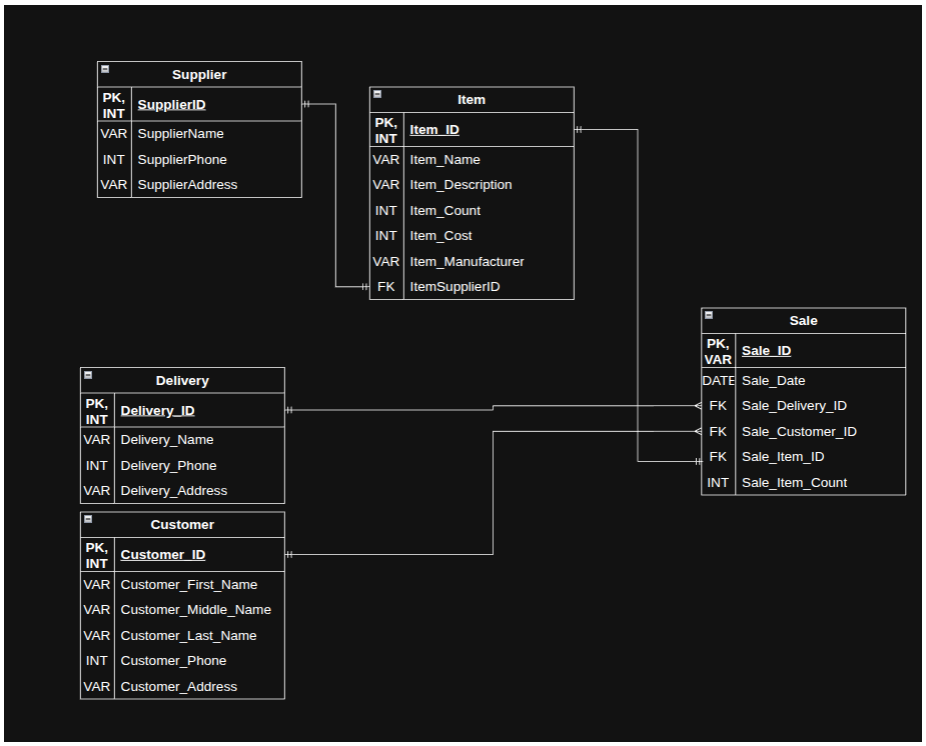
Supplier
PK, INT SupplierID
VAR SupplierName
INT SupplierPhone
VAR SupplierAddress
Item
PK, INT Item_ID
VAR Item_Name
VAR Item_Description
INT Item_Count
INT Item_Cost
VAR Item_Manufacturer
FK ItemSupplierID
Sale
PK, VAR Sale_ID
DATE Sale_Date
FK Sale_Delivery_ID
FK Sale_Customer_ID
FK Sale_Item_ID
INT Sale_Item_Count
Delivery
PK, INT Delivery_ID
VAR Delivery_Name
INT Delivery_Phone
VAR Delivery_Address
Customer
PK, INT Customer_ID
VAR Customer_First_Name
VAR Customer_Middle_Name
VAR Customer_Last_Name
INT Customer_Phone
VAR Customer_Address

Figure 1: img

Address for Customer, Supplier and Delivery have to be atomized

- Example 2: Item Foreign Key of Sale Table -> Seperated into A different table: Receipt Table containing ItemID, SaleID Forein Keys and Item Count Property to allow the user to buy multiple different items in one sale.
- End result:
  * Customer:
    · CustomerID (INT Primary Key)
    · CustomerFirstName (VARCHAR)
    · CustomerSecondName (VARCHAR)
    · CustomerLastName (VARCHAR)
    · CustomerPhone (INT)
    · CustomerAddress (VARCHAR)
    · CustomerStreet (VARCHAR)
    · CustomerZip (VARCHAR)
    · CustomerCountry (VARCHAR)
    · CustomerRegion (VARCHAR)
    · CustomerCity (VARCHAR)
  * Delivery:
    · DeliveryID (INT Primary Key)
    · DeliveryName (VARCHAR)
    · DeliveryFirstName (VARCHAR)
    · DeliverySecondName (VARCHAR)
    · DeliveryLastName (VARCHAR)
    · DeliveryPhone (INT)
    · DeliveryAddress (VARCHAR)
    · DeliveryStreet (VARCHAR)
    · DeliveryZip (VARCHAR)
    · DeliveryCountry (VARCHAR)
    · DeliveryRegion (VARCHAR)
    · DeliveryCity (VARCHAR)
  * Supplier:
    · SupplierID (INT Primary Key)
    · SupplierName (VARCHAR)
    · SupplierFirstName (VARCHAR)
    · SupplierSecondName (VARCHAR)
    · SupplierLastName (VARCHAR)
    · SupplierPhone (INT)
    · SupplierAddress (VARCHAR)
    · SupplierStreet (VARCHAR)
    · SupplierZip (VARCHAR)
    · SupplierCountry (VARCHAR)
    · SupplierRegion (VARCHAR)
    · SupplierCity (VARCHAR)
  * Sale:
    · SaleID (INT Primary Key)

        - SaleDeliveryID (INT Foreign Key)
        - SaleCustomerID (INT Foreign Key)
    - Receipt:
        - ReceiptSaleID (INT Foreign Key)
        - ReceiptItemID (INT Foreign Key)
        - ReceiptItemCount (INT)
- 1NF to 2NF: Non-Key attribute must depend on every part of the primary Key
    - All values of the tables abide by the Normalization, namely the Receipt Table using compost Primary Key structure (two or more foreign keys to identify a table)
- 2NF to 3NF: Non-key attributes not depending on eachother
    - Addresses of three aforementioned tables are dependent on eachother and is not compact
    - Solution: Seperating Addresses into individual Address tables for 3 entities, pointing to a seperate Zip Code table, the Zip table containing values that can be determined from the zip code, and addresses contain the rest
        - Zip:
            - ZipCode (VARCHAR Primary Key)
            - Country (VARCHAR)
            - Region (VARCHAR)
            - City (VARCHAR)
        - CustomerAddress:
            - CustomerAddressID (INT Primary Key)
            - CustomerAddressStreet (INT Primary Key)
            - CustomerAddressStreetNum (INT Primary Key)
            - CustomerAddressZip (VARCHAR Foreign Key)
        - DeliveryAddress:
            - DeliveryAddressID (INT Primary Key)
            - DeliveryAddressStreet (INT Primary Key)
            - DeliveryAddressStreetNum (INT Primary Key)
            - DeliveryAddressZip (VARCHAR Foreign Key)
        - SupplierAddress:
            - SupplierAddressID (INT Primary Key)
            - SupplierAddressStreet (INT Primary Key)
            - SupplierAddressStreetNum (INT Primary Key)
            - SupplierAddressZip (VARCHAR Foreign Key)
        - Customer:
            - CustomerID (INT Primary Key)
            - CustomerFirstName (VARCHAR)
            - CustomerSecondName (VARCHAR)
            - CustomerLastName (VARCHAR)
            - CustomerPhone (INT)
            - CustomerAddressIDCode (INT Foreign Key)
        - Delivery:

- DeliveryID (INT Primary Key)
- DeliveryName (VARCHAR)
- DeliveryFirstName (VARCHAR)
- DeliverySecondName (VARCHAR)
- DeliveryLastName (VARCHAR)
- DeliveryPhone (INT)
- DeliveryAddressIDCode (INT Foreign Key)

* Supplier:
  - SupplierID (INT Primary Key)
  - SupplierName (VARCHAR)
  - SupplierFirstName (VARCHAR)
  - SupplierSecondName (VARCHAR)
  - SupplierLastName (VARCHAR)
  - SupplierPhone (INT)
  - SupplierAddressIDCode (INT Foreign Key)

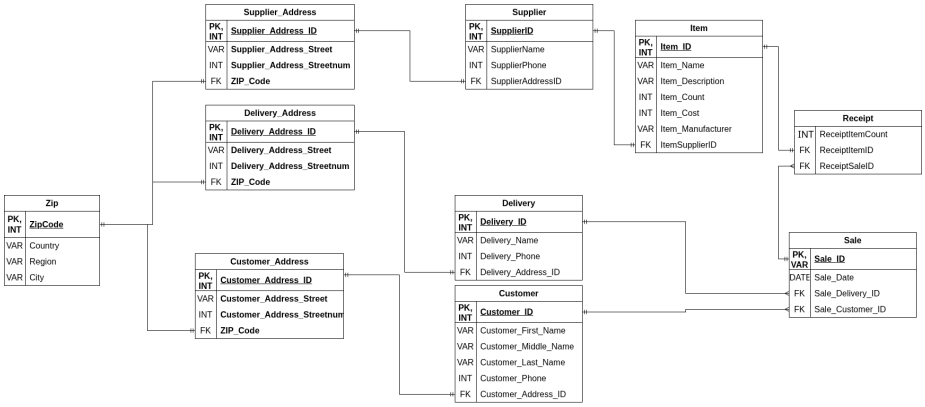

Figure 2: img

Requirement d:

Source code for the table:

```
CREATE TABLE Zip
(
    ZipCode VARCHAR(128) NOT NULL PRIMARY KEY,
    Country VARCHAR(128),
    Region VARCHAR(128),
    City VARCHAR(128)1
);

CREATE TABLE CustomerAddress
(
    CustomerAddressID INT NOT NULL PRIMARY KEY,
```

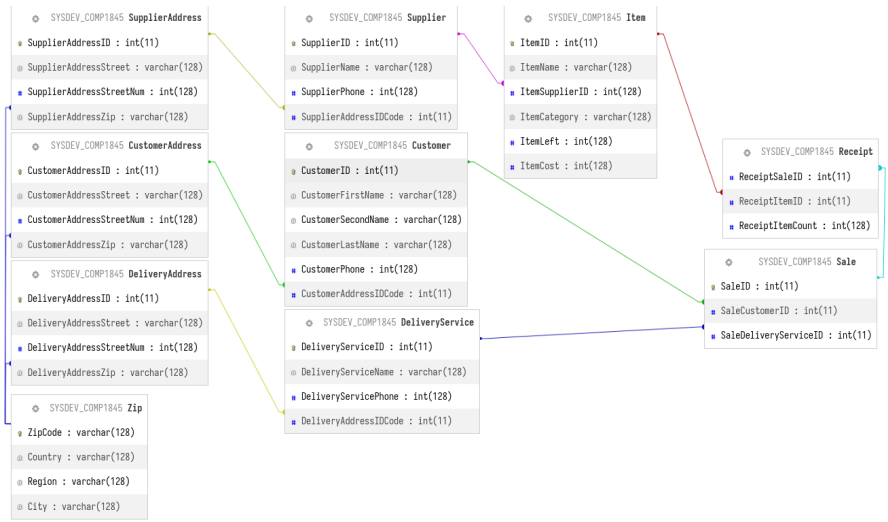

Figure 3: img

	#	Name	Type	Collation	Attributes	Null	Default	Comments	Extra	Action		
☐	1	ZipCode	varchar(128)	*utf8mb4_general_ci*		No	*None*			Change	Drop	More
☐	2	Country	varchar(128)	*utf8mb4_general_ci*		Yes	*NULL*			Change	Drop	More
☐	3	Region	varchar(128)	*utf8mb4_general_ci*		Yes	*NULL*			Change	Drop	More
☐	4	City	varchar(128)	*utf8mb4_general_ci*		Yes	*NULL*			Change	Drop	More

Figure 4: img

```
    CustomerAddressStreet VARCHAR(128),
    CustomerAddressStreetNum INT(128),
    CustomerAddressZip VARCHAR(128),
    FOREIGN KEY (CustomerAddressZip) REFERENCES Zip(ZipCode)
);
```

#	Name	Type	Collation	Attributes	Null	Default	Comments	Extra	Action		
1	CustomerAddressID	int(11)			No	None			Change	Drop	More
2	CustomerAddressStreet	varchar(128)	utf8mb4_general_ci		Yes	NULL			Change	Drop	More
3	CustomerAddressStreetNum	int(128)			Yes	NULL			Change	Drop	More
4	CustomerAddressZip	varchar(128)	utf8mb4_general_ci		Yes	NULL			Change	Drop	More

Figure 5: img

```
CREATE TABLE DeliveryAddress
(
    DeliveryAddressID INT NOT NULL PRIMARY KEY,
    DeliveryAddressStreet VARCHAR(128),
    DeliveryAddressStreetNum INT(128),
    DeliveryAddressZip VARCHAR(128),
    FOREIGN KEY (DeliveryAddressZip) REFERENCES Zip(ZipCode)
);
```

#	Name	Type	Collation	Attributes	Null	Default	Comments	Extra	Action		
1	DeliveryAddressID	int(11)			No	None			Change	Drop	More
2	DeliveryAddressStreet	varchar(128)	utf8mb4_general_ci		Yes	NULL			Change	Drop	More
3	DeliveryAddressStreetNum	int(128)			Yes	NULL			Change	Drop	More
4	DeliveryAddressZip	varchar(128)	utf8mb4_general_ci		Yes	NULL			Change	Drop	More

Figure 6: img

```
CREATE TABLE SupplierAddress
(
    SupplierAddressID INT NOT NULL PRIMARY KEY,
    SupplierAddressStreet VARCHAR(128),
    SupplierAddressStreetNum INT(128),
    SupplierAddressZip VARCHAR(128),
    FOREIGN KEY (SupplierAddressZip) REFERENCES Zip(ZipCode)
);
```

```
CREATE TABLE Customer
(
    CustomerID INT NOT NULL PRIMARY KEY,
    CustomerFirstName VARCHAR(128),
    CustomerSecondName VARCHAR(128),
```

#	Name	Type	Collation	Attributes	Null	Default	Comments	Extra	Action		
1	SupplierAddressID	int(11)			No	None			Change	Drop	More
2	SupplierAddressStreet	varchar(128)	utf8mb4_general_ci		Yes	NULL			Change	Drop	More
3	SupplierAddressStreetNum	int(128)			Yes	NULL			Change	Drop	More
4	SupplierAddressZip	varchar(128)	utf8mb4_general_ci		Yes	NULL			Change	Drop	More

Figure 7: img

```
    CustomerLastName VARCHAR(128),
    CustomerPhone INT(128),
    CustomerAddressIDCode INT,
    FOREIGN KEY (CustomerAddressIDCode) REFERENCES CustomerAddress(CustomerAddressID)
);
```

#	Name	Type	Collation	Attributes	Null	Default	Comments	Extra	Action		
1	CustomerID	int(11)			No	None			Change	Drop	More
2	CustomerFirstName	varchar(128)	utf8mb4_general_ci		Yes	NULL			Change	Drop	More
3	CustomerSecondName	varchar(128)	utf8mb4_general_ci		Yes	NULL			Change	Drop	More
4	CustomerLastName	varchar(128)	utf8mb4_general_ci		Yes	NULL			Change	Drop	More
5	CustomerPhone	int(128)			Yes	NULL			Change	Drop	More
6	CustomerAddressIDCode	int(11)			Yes	NULL			Change	Drop	More

Figure 8: img

```
CREATE TABLE Supplier
(
    SupplierID INT NOT NULL PRIMARY KEY,
    SupplierName VARCHAR(128),
    SupplierPhone INT(128),
    SupplierAddressIDCode INT,
    FOREIGN KEY (SupplierAddressIDCode) REFERENCES SupplierAddress(SupplierAddressID)
);
```

#	Name	Type	Collation	Attributes	Null	Default	Comments	Extra	Action		
1	SupplierID	int(11)			No	None			Change	Drop	More
2	SupplierName	varchar(128)	utf8mb4_general_ci		Yes	NULL			Change	Drop	More
3	SupplierPhone	int(128)			Yes	NULL			Change	Drop	More
4	SupplierAddressIDCode	int(11)			Yes	NULL			Change	Drop	More

Figure 9: img

```
CREATE TABLE Item
(
```

```
    ItemID INT NOT NULL PRIMARY KEY,
    ItemName VARCHAR(128),
    ItemSupplierID INT(128),
    ItemCategory VARCHAR(128),
    ItemLeft INT(128),
    ItemCost INT(128),
    CONSTRAINT SupplierID FOREIGN KEY (ItemSupplierID)
    REFERENCES Supplier(SupplierID)
);
```

#	Name	Type	Collation	Attributes	Null	Default	Comments	Extra	Action		
1	ItemID	int(11)			No	*None*			Change	Drop	More
2	ItemName	varchar(128)	*utf8mb4_general_ci*		Yes	*NULL*			Change	Drop	More
3	ItemSupplierID	int(128)			Yes	*NULL*			Change	Drop	More
4	ItemCategory	varchar(128)	*utf8mb4_general_ci*		Yes	*NULL*			Change	Drop	More
5	ItemLeft	int(128)			Yes	*NULL*			Change	Drop	More
6	ItemCost	int(128)			Yes	*NULL*			Change	Drop	More

Figure 10: img

```
CREATE TABLE DeliveryService
(
    DeliveryServiceID INT NOT NULL PRIMARY KEY,
    DeliveryServiceName VARCHAR(128),
    DeliveryServicePhone INT(128),
    DeliveryAddressIDCode INT,
    FOREIGN KEY (DeliveryAddressIDCode) REFERENCES DeliveryAddress(DeliveryAddressID)
);
```

#	Name	Type	Collation	Attributes	Null	Default	Comments	Extra	Action		
1	DeliveryServiceID	int(11)			No	*None*			Change	Drop	More
2	DeliveryServiceName	varchar(128)	*utf8mb4_general_ci*		Yes	*NULL*			Change	Drop	More
3	DeliveryServicePhone	int(128)			Yes	*NULL*			Change	Drop	More
4	DeliveryAddressIDCode	int(11)			Yes	*NULL*			Change	Drop	More

Figure 11: img

```
CREATE TABLE Sale
(
    SaleID INT NOT NULL PRIMARY KEY,
    SaleCustomerID INT,
    SaleDeliveryServiceID INT,
    FOREIGN KEY (SaleCustomerID) REFERENCES Customer(CustomerID),
    FOREIGN KEY (SaleDeliveryServiceID) REFERENCES DeliveryService(DeliveryServiceID)
```

```
);
```

#	Name	Type	Collation	Attributes	Null	Default	Comments	Extra	Action		
1	SaleID	int(11)			No	None			Change	Drop	More
2	SaleCustomerID	int(11)			Yes	NULL			Change	Drop	More
3	SaleDeliveryServiceID	int(11)			Yes	NULL			Change	Drop	More

Figure 12: img

```
CREATE TABLE Receipt
(
    ReceiptSaleID INT,
    ReceiptItemID INT,
    ReceiptItemCount INT(128),
    FOREIGN KEY (ReceiptSaleID) REFERENCES Sale(SaleID),
    FOREIGN KEY (ReceiptItemID) REFERENCES Item(ItemID)
);
```

#	Name	Type	Collation	Attributes	Null	Default	Comments	Extra	Action		
1	ReceiptSaleID	int(11)			Yes	NULL			Change	Drop	More
2	ReceiptItemID	int(11)			Yes	NULL			Change	Drop	More
3	ReceiptItemCount	int(128)			Yes	NULL			Change	Drop	More

Figure 13: img

Requirement e:

```
INSERT INTO Zip (ZipCode, Country, Region, City) VALUES
('10001', 'United States', 'New York', 'New York'),
('90210', 'United States', 'California', 'Beverly Hills'),
('60601', 'United States', 'Illinois', 'Chicago'),
('33101', 'United States', 'Florida', 'Miami'),
('75201', 'United States', 'Texas', 'Dallas'),
('94102', 'United States', 'California', 'San Francisco'),
('02108', 'United States', 'Massachusetts', 'Boston'),
('80202', 'United States', 'Colorado', 'Denver'),
('98101', 'United States', 'Washington', 'Seattle'),
('30301', 'United States', 'Georgia', 'Atlanta'),
('2000', 'Australia', 'New South Wales', 'Sydney'),
('3000', 'Australia', 'Victoria', 'Melbourne'),
('4000', 'Australia', 'Queensland', 'Brisbane'),
('5000', 'Australia', 'South Australia', 'Adelaide'),
('6000', 'Australia', 'Western Australia', 'Perth'),
```

```
('85001', 'United States', 'Arizona', 'Phoenix'),
('19102', 'United States', 'Pennsylvania', 'Philadelphia'),
('43201', 'United States', 'Ohio', 'Columbus'),
('48201', 'United States', 'Michigan', 'Detroit'),
('20001', 'United States', 'District of Columbia', 'Washington'),
('70001', 'United States', 'Louisiana', 'Metairie'),
('89101', 'United States', 'Nevada', 'Las Vegas'),
('97201', 'United States', 'Oregon', 'Portland'),
('37201', 'United States', 'Tennessee', 'Nashville'),
('55401', 'United States', 'Minnesota', 'Minneapolis'),
('2600', 'Australia', 'ACT', 'Canberra'),
('7000', 'Australia', 'Tasmania', 'Hobart'),
('0800', 'Australia', 'Northern Territory', 'Darwin'),
('64101', 'United States', 'Missouri', 'Kansas City'),
('46201', 'United States', 'Indiana', 'Indianapolis');

INSERT INTO CustomerAddress (CustomerAddressID, CustomerAddressStreet, CustomerAddressStreet
(482930, 'Rodeo Drive', 421, '90210'),
(102938, 'Canon Drive', 712, '90210'),
(556473, 'Lomitas Avenue', 1105, '90210'),
(827364, 'Alpine Drive', 804, '90210'),
(192837, 'Bedford Drive', 516, '90210'),
(475869, 'Camden Drive', 625, '90210'),
(334455, 'Elevado Avenue', 902, '90210'),
(998877, 'Foothill Road', 1240, '90210'),
(221133, 'Linden Drive', 709, '90210'),
(665544, 'Maple Drive', 815, '90210'),
(774411, 'Palm Drive', 503, '90210'),
(123546, 'Rexford Drive', 618, '90210'),
(889900, 'Sierra Drive', 722, '90210'),
(445566, 'Walden Drive', 935, '90210'),
(112233, 'Carmelita Avenue', 1010, '90210'),
(990011, 'Beverly Drive', 455, '90210'),
(332211, 'Crescent Drive', 312, '90210'),
(554433, 'Dayton Way', 9440, '90210'),
(778899, 'Brighton Way', 9520, '90210'),
(667788, 'Alta Drive', 801, '90210'),
(223344, 'Arden Drive', 606, '90210'),
(446688, 'Hillcrest Road', 1175, '90210'),
(101020, 'Mountain Drive', 1005, '90210'),
(302010, 'Sunset Boulevard', 9641, '90210'),
(506070, 'Benedict Canyon Drive', 2500, '90210'),
(807060, 'Tower Road', 1120, '90210'),
(905040, 'Coldwater Canyon Drive', 1405, '90210'),
(204060, 'Lexington Road', 812, '90210'),
(705030, 'Greenway Drive', 901, '90210'),
```

```
(105090, 'Oxford Way', 1215, '90210');
(582914, 'Via Dante', 14, '20121');

INSERT INTO DeliveryAddress (DeliveryAddressID, DeliveryAddressStreet, DeliveryAddressStreet
(293847, 'Loma Vista Drive', 1125, '90210'),
(847561, 'Doheny Road', 501, '90210'),
(102948, 'Bowmont Drive', 2410, '90210'),
(667382, 'Deep Canyon Drive', 1805, '90210'),
(550493, 'Wallace Ridge', 1300, '90210'),
(384756, 'Carla Ridge', 1185, '90210'),
(928374, 'Schuyler Road', 912, '90210'),
(475610, 'Robert Lane', 1004, '90210'),
(119283, 'Trousdale Estates', 425, '90210'),
(663322, 'Chalette Drive', 510, '90210'),
(774859, 'Arkell Drive', 1401, '90210'),
(203948, 'Lania Lane', 1210, '90210'),
(884756, 'Summit Drive', 1100, '90210'),
(449302, 'Pickfair Way', 1143, '90210'),
(556677, 'Angelo Drive', 1280, '90210'),
(123847, 'Brooktree Road', 605, '90210'),
(904857, 'Heather Road', 955, '90210'),
(334411, 'Dutton Drive', 814, '90210'),
(775599, 'Hidden Valley Road', 2505, '90210'),
(662288, 'Franklin Canyon Drive', 1900, '90210'),
(229944, 'Kimridge Road', 1420, '90210'),
(441188, 'Claridge Drive', 1015, '90210'),
(109020, 'San Ysidro Drive', 1602, '90210'),
(308010, 'Tower Grove Drive', 1220, '90210'),
(502070, 'Waring Avenue', 445, '90210'),
(801060, 'Gregory Way', 315, '90210'),
(902040, 'Charleville Boulevard', 9420, '90210'),
(203060, 'Lasky Drive', 105, '90210'),
(701030, 'Spalding Drive', 222, '90210'),
(107090, 'Oakhurst Drive', 418, '90210');

INSERT INTO SupplierAddress (SupplierAddressID, SupplierAddressStreet, SupplierAddressStreet
(314256, 'Carla Lane', 1018, '90210'),
(827319, 'Bundy Drive', 1540, '90210'),
(665102, 'Ambassador Avenue', 1001, '90210'),
(449281, 'Shadow Hill Way', 1225, '90210'),
(510293, 'Wallace Ridge', 1350, '90210'),
(883746, 'Dona Marina Drive', 3105, '90210'),
(192830, 'Deep Canyon Drive', 2112, '90210'),
(774123, 'Coldwater Canyon Drive', 1845, '90210'),
(556201, 'Kimridge Road', 1390, '90210'),
(304928, 'Bowmont Drive', 2555, '90210'),
```

```
(495817, 'Lania Lane', 1288, '90210'),
(202134, 'San Ysidro Drive', 1701, '90210'),
(993847, 'Tower Grove Drive', 1140, '90210'),
(440011, 'Heather Road', 9210, '90210'),
(667123, 'Hidden Valley Road', 2480, '90210'),
(129485, 'Cielo Drive', 10050, '90210'),
(338271, 'Benedict Canyon Drive', 2615, '90210'),
(772109, 'Hutton Drive', 1905, '90210'),
(884102, 'Beverly Estate Drive', 1201, '90210'),
(559182, 'Loma Vista Drive', 1195, '90210'),
(443322, 'Doheny Road', 412, '90210'),
(221199, 'Sierra Drive', 802, '90210'),
(664433, 'Palm Drive', 615, '90210'),
(779900, 'Maple Drive', 710, '90210'),
(102030, 'Linden Drive', 505, '90210'),
(405060, 'Roxbury Drive', 915, '90210'),
(809010, 'Bedford Drive', 601, '90210'),
(204080, 'Peck Drive', 335, '90210'),
(706050, 'Canon Drive', 422, '90210'),
(901020, 'Beverly Drive', 210, '90210');

INSERT INTO Supplier (SupplierID, SupplierName, SupplierPhone, SupplierAddressIDCode) VALUES
(482910, 'Apex Logistics Solutions', 31550192, 556201),
(102938, 'Golden Gate Wholesale', 21358821, 556201),
(556473, 'Summit Manufacturing', 32355732, 556201),
(827364, 'Pacific Rim Imports', 41555012, 556201),
(192837, 'Blue Horizon Textiles', 81852345, 556201),
(475869, 'Beverly Hills Tech Corp', 31557766, 556201),
(334455, 'Emerald City Trading', 20655122, 556201),
(998877, 'Starlight Distribution', 71558899, 556201),
(221133, 'Precision Parts Inc', 61955344, 556201),
(665544, 'Coastal Supply Co', 80555577, 556201),
(774411, 'NextGen Electronics', 40850011, 556201),
(123546, 'Global Reach Freight', 51052233, 556201),
(889900, 'Mainland Materials', 91655455, 556201),
(112233, 'North Star Industrial', 61257788, 556201),
(990011, 'Desert Springs Water', 60255500, 556201),
(332211, 'Empire State Sourcing', 21251100, 556201),
(554433, 'Windy City Hardware', 31252299, 556201),
(778899, 'Lone Star Steel', 51255388, 556201),
(667788, 'Peach State Produce', 45554477, 556201),
(223344, 'Evergreen Forestry', 55555566, 556201),
(446688, 'Bayou Beverage Group', 55556655, 556201),
(101020, 'Silver State Mining', 75557744, 556201),
(302010, 'Cascade Equipment', 25558833, 556201),
(506070, 'Sunshine State Citrus', 55559922, 556201),
```

```
(807060, 'Rocky Mountain Gear', 35551010, 556201),
(905040, 'Gateway City Freight', 45552020, 556201),
(204060, 'Liberty Bell Fabrics', 55553030, 556201),
(705030, 'Old Dominion Paper', 45554040, 556201),
(105090, 'Great Lakes Glass', 35555050, 556201),
(339922, 'Canyon Rim Partners', 15556060, 556201);

INSERT INTO DeliveryService (DeliveryServiceID, DeliveryServiceName, DeliveryServicePhone, D
(219485, 'SwiftStream Logistics', 31051029, 123847),
(774821, 'Velox Parcel Services', 21359930, 123847),
(550192, 'HyperLink Couriers', 32355582, 123847),
(837261, 'DirectRoute Freight', 41555721, 123847),
(192038, 'Blue Dart Deliveries', 81853344, 123847),
(476102, '90210 Express Mail', 31055510, 123847),
(339485, 'City-to-City Haulage', 20651212, 123847),
(991122, 'NextDay Global Trans', 71454433, 123847),
(228374, 'Point-B Logistics', 61955509, 123847),
(661029, 'Pacific West Cargo', 80555561, 123847),
(775533, 'RapidResponse Dispatch', 40552020, 123847),
(120394, 'Gold Star Trucking', 51055531, 123847),
(884712, 'MileHigh Courier Co', 91655242, 123847),
(119933, 'Interstate Expeditors', 61255353, 123847),
(998811, 'Red Zone Transport', 60255564, 123847),
(331100, 'IronClad Shipping', 21255575, 123847),
(552288, 'Windy City Couriers', 31255686, 123847),
(773399, 'Lone Star Logistics', 51255797, 123847),
(664477, 'Southern Cross Freight', 40550808, 123847),
(225544, 'Northwest Cargo Link', 50351919, 123847),
(447766, 'Bayou Rapid Transit', 50455030, 123847),
(108822, 'Silver State Express', 70254141, 123847),
(301144, 'Peak Performance Mail', 25355252, 123847),
(509977, 'Sunshine State Deliver', 30556363, 123847),
(802266, 'Alpine Logistics Group', 30557474, 123847),
(904433, 'Midwest Haulage Co', 31455585, 123847),
(206655, 'Liberty Bell Logistics', 21559696, 123847),
(708844, 'Commonwealth Cargo', 80455507, 123847),
(103366, 'Great Lakes Dispatch', 31351818, 123847),
(337711, 'Canyon View Couriers', 80152929, 123847);

INSERT INTO Customer (CustomerID, CustomerFirstName, CustomerSecondName, CustomerLastName, C
(829401, 'Giovanni', 'Battista', 'Moretti', 390555301, 582914),
(789234, 'Barry', 'The', 'Bot', 720137410, 582914),
(104928, 'Antonio', 'Pasquale', 'Esposito', 390555302, 582914),
(556102, 'Vincenzo', 'Domenico', 'Ricci', 390255303, 582914),
(339481, 'Salvatore', 'Angelo', 'Marino', 390255304, 582914),
(771122, 'Stefano', 'Roberto', 'Greco', 390255305, 582914),
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(228375, 'Giuseppe', 'Franco', 'Bruno', 390255306, 582914),
(661030, 'Claudio', 'Massimo', 'Gallo', 390255307, 582914),
(775534, 'Daniele', 'Enrico', 'Conti', 390255508, 582914),
(120395, 'Fabrizio', 'Paolo', 'De Luca', 390255309, 582914),
(884713, 'Gianluca', 'Sandro', 'Mancini', 390255310, 582914),
(119934, 'Carmine', 'Renato', 'Costa', 390255311, 582914),
(998812, 'Pasquale', 'Gino', 'Giordano', 39025512, 582914),
(331101, 'Giorgio', 'Filippo', 'Rizzo', 390255513, 582914),
(552289, 'Raffaele', 'Umberto', 'Lombardi', 392555314, 582914),
(773398, 'Pietro', 'Rosario', 'Barbieri', 390555315, 582914),
(664476, 'Emilio', 'Cesare', 'Fontana', 390255316, 582914),
(225545, 'Augusto', 'Tullio', 'Santoro', 392555317, 582914),
(447767, 'Mauro', 'Silvio', 'Mariani', 390555318, 582914),
(108823, 'Flavio', 'Valerio', 'Rinaldi', 390255319, 582914),
(301145, 'Giacomo', 'Ettore', 'Caruso', 390255320, 582914),
(509978, 'Luciano', 'Nico', 'Ferraro', 390255321, 582914),
(802267, 'Enzo', 'Ottavio', 'Valli', 390255522, 582914),
(904434, 'Paolo', 'Santi', 'Guerra', 390255323, 582914),
(206656, 'Renzo', 'Ivano', 'Pellegrini', 302555324, 582914),
(708845, 'Aldo', 'Nino', 'Sanna', 390255525, 582914),
(103367, 'Bruno', 'Alfonso', 'Farina', 302555326, 582914),
(337712, 'Carlo', 'Arturo', 'Ferrero', 902555327, 582914),
(441122, 'Dario', 'Duccio', 'Serra', 390255328, 582914),
(558899, 'Fausto', 'Lio', 'Vitale', 390255329, 582914),
(663300, 'Igor', 'Zeno', 'Messina', 390555330, 582914);

INSERT INTO Item (ItemID, ItemName, ItemSupplierID, ItemCategory, ItemLeft, ItemCost) VALUES
(592831, 'Silk Egyptian Sheets', 112233, 'Bedding', 142, 450),
(102948, 'Smart Espresso Machine', 112233, 'Appliances', 45, 1200),
(884729, 'Teak Outdoor Lounge', 112233, 'Furniture', 12, 3400),
(334102, 'Crystal Stemware Set', 112233, 'Kitchenware', 89, 250),
(771029, 'Noise-Cancelling Headphones', 112233, 'Electronics', 230, 350),
(550192, 'Cashmere Throw Blanket', 112233, 'Home Decor', 67, 180),
(219384, 'Professional Chef Knife', 112233, 'Kitchenware', 156, 125),
(667102, 'Velvet Occasional Chair', 112233, 'Furniture', 8, 890),
(440293, 'Smart Home Hub', 112233, 'Electronics', 412, 199),
(119283, 'Aromatic Soy Candle', 112233, 'Home Decor', 850, 45),
(903847, 'Electric Stand Mixer', 112233, 'Appliances', 34, 550),
(482716, 'Goose Down Duvet', 112233, 'Bedding', 22, 600),
(773322, 'Ergonomic Office Chair', 112233, 'Furniture', 56, 1100),
(120394, 'Bluetooth Surround Bar', 112233, 'Electronics', 115, 800),
(884712, 'Cast Iron Dutch Oven', 112233, 'Kitchenware', 94, 280),
(556611, 'Organic Cotton Towels', 112233, 'Bath', 540, 75),
(339485, 'Dimmable Floor Lamp', 112233, 'Lighting', 43, 320),
(991122, 'Porcelain Dinnerware Set', 112233, 'Kitchenware', 31, 450),
(228374, 'Stainless Steel Air Fryer', 112233, 'Appliances', 128, 220),
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(661029, 'Memory Foam Mattress', 112233, 'Bedding', 14, 2500),
(775533, 'Smart Door Lock', 112233, 'Electronics', 204, 299),
(120395, 'Hand-Woven Area Rug', 112233, 'Home Decor', 9, 4200),
(884713, 'Copper Cookware Set', 112233, 'Kitchenware', 18, 1500),
(119934, 'Air Purifying System', 112233, 'Appliances', 65, 450),
(998811, 'Abstract Oil Painting', 112233, 'Home Decor', 5, 4800),
(331100, 'Adjustable Standing Desk', 112233, 'Furniture', 27, 1350),
(552288, 'Cordless Stick Vacuum', 112233, 'Appliances', 82, 700),
(773399, 'Smart Garden Planter', 112233, 'Garden', 150, 160),
(664477, 'Crystal Chandelier', 112233, 'Lighting', 3, 4950),
(225544, 'Wine Cellar Cooling Unit', 112233, 'Appliances', 7, 3200);

INSERT INTO Sale (SaleID, SaleCustomerID, SaleDeliveryServiceID) VALUES
(192837, 108823, 331100),
(475869, 108823, 331100),
(334455, 108823, 331100),
(998877, 108823, 331100),
(221133, 108823, 331100),
(665544, 108823, 331100),
(774411, 108823, 331100),
(123546, 108823, 331100),
(889900, 108823, 331100),
(112233, 108823, 331100),
(990011, 108823, 331100),
(332211, 108823, 331100),
(554433, 108823, 331100),
(778899, 108823, 331100),
(667788, 108823, 331100),
(223344, 108823, 331100),
(446688, 108823, 331100),
(101020, 108823, 331100),
(302010, 108823, 331100),
(506070, 108823, 331100),
(807060, 108823, 331100),
(905040, 108823, 331100),
(204060, 108823, 331100),
(705030, 108823, 331100),
(105090, 108823, 331100),
(339922, 108823, 331100),
(415263, 108823, 331100),
(987654, 108823, 331100),
(123456, 108823, 331100),
(654321, 108823, 331100);

INSERT INTO Receipt (ReceiptSaleID, ReceiptItemID, ReceiptItemCount) VALUES
(665544, 592831, 2),
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(665544, 102948, 1),
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(665544, 334102, 5),
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(665544, 550192, 3),
(665544, 219384, 1),
(665544, 667102, 1),
(665544, 440293, 5),
(665544, 119283, 4),
(665544, 903847, 1),
(665544, 482716, 2),
(665544, 773322, 3),
(665544, 120394, 1),
(665544, 884712, 5),
(665544, 556611, 4),
(665544, 339485, 2),
(665544, 991122, 3),
(665544, 228374, 1),
(665544, 661029, 2),
(665544, 775533, 5),
(665544, 120395, 1),
(665544, 884713, 2),
(665544, 119934, 1),
(665544, 998811, 1),
(665544, 331100, 3),
(665544, 552288, 1),
(665544, 773399, 4),
(665544, 664477, 1),
(665544, 225544, 1);
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Screenshots:

CustomerID	CustomerFirstName	CustomerSecondName	CustomerLastName	CustomerPhone	CustomerAddressIDCode
103367	Bruno	Alfonso	Farina	302555326	582914
104928	Antonio	Pasquale	Esposito	390555302	582914
108823	Flavio	Valerio	Rinaldi	390255319	582914
119934	Carmine	Renato	Costa	390255311	582914
120395	Fabrizio	Paolo	De Luca	390255309	582914
206656	Renzo	Ivano	Pellegrini	302555324	582914
225545	Augusto	Tullio	Santoro	392555317	582914
228375	Giuseppe	Franco	Bruno	390255306	582914
301145	Giacomo	Ettore	Caruso	390255320	582914
331101	Giorgio	Filippo	Rizzo	390255513	582914
337712	Carlo	Arturo	Ferrero	902555327	582914
339481	Salvatore	Angelo	Marino	390255304	582914
441122	Dario	Duccio	Serra	390255328	582914
447767	Mauro	Silvio	Mariani	390555318	582914
509978	Luciano	Nico	Ferraro	390255321	582914
552289	Raffaele	Umberto	Lombardi	392555314	582914
556102	Vincenzo	Domenico	Ricci	390255303	582914
558899	Fausto	Lio	Vitale	390255329	582914
661030	Claudio	Massimo	Gallo	390255307	582914
663300	Igor	Zeno	Messina	390555330	582914
664476	Emilio	Cesare	Fontana	390255316	582914
708845	Aldo	Nino	Sanna	390255525	582914
771122	Stefano	Roberto	Greco	390255305	582914
773398	Pietro	Rosario	Barbieri	390555315	582914
775534	Daniele	Enrico	Conti	390255508	582914

CustomerID	CustomerFirstName	CustomerSecondName	CustomerLastName	CustomerPhone	CustomerAddressIDCode
789234	Barry	The	Bot	720137410	582914
802267	Enzo	Ottavio	Valli	390255522	582914
827364	Alessandro	Pietro	Romano	392555205	550493
829401	Giovanni	Battista	Moretti	390555301	582914
884713	Gianluca	Sandro	Mancini	390255310	582914
889900	Ludovica	Vittoria	Costa	390255214	449302
904434	Paolo	Santi	Guerra	390255323	582914
905040	Manuel	Nicolas	Guerra	390255227	902040
918273	Giovanni	Battista	Moretti	392555231	582914
990011	Nicole	Noemi	Rizzo	390255216	123847
998812	Pasquale	Gino	Giordano	39025512	582914
998877	Mattia	Davide	Bruno	39025509	119283

CustomerAddressID	CustomerAddressStreet	CustomerAddressStreetNum	CustomerAddressZip
101020	Mountain Drive	1005	90210
102938	Canon Drive	712	90210
105090	Oxford Way	1215	90210
112233	Carmelita Avenue	1010	90210
123546	Rexford Drive	618	90210
192837	Bedford Drive	516	90210
204060	Lexington Road	812	90210
221133	Linden Drive	709	90210
223344	Arden Drive	606	90210
302010	Sunset Boulevard	9641	90210
332211	Crescent Drive	312	90210
334455	Elevado Avenue	902	90210
445566	Walden Drive	935	90210
446688	Hillcrest Road	1175	90210
475869	Camden Drive	625	90210
482930	Rodeo Drive	421	90210
506070	Benedict Canyon Drive	2500	90210
554433	Dayton Way	9440	90210
556473	Lomitas Avenue	1105	90210
582914	Via Dante	14	20121
665544	Maple Drive	815	90210
667788	Alta Drive	801	90210
705030	Greenway Drive	901	90210
774411	Palm Drive	503	90210
778899	Brighton Way	9520	90210

CustomerAddressID	CustomerAddressStreet	CustomerAddressStreetNum	CustomerAddressZip
807060	Tower Road	1120	90210
827364	Alpine Drive	804	90210
889900	Sierra Drive	722	90210
905040	Coldwater Canyon Drive	1405	90210
990011	Beverly Drive	455	90210
998877	Foothill Road	1240	90210

DeliveryAddressID	DeliveryAddressStreet	DeliveryAddressStreetNum	DeliveryAddressZip
102948	Bowmont Drive	2410	90210
107090	Oakhurst Drive	418	90210
109020	San Ysidro Drive	1602	90210
119283	Trousdale Estates	425	90210
123847	Brooktree Road	605	90210
203060	Lasky Drive	105	90210
203948	Lania Lane	1210	90210
229944	Kimridge Road	1420	90210
293847	Loma Vista Drive	1125	90210
308010	Tower Grove Drive	1220	90210
334411	Dutton Drive	814	90210
384756	Carla Ridge	1185	90210
441188	Claridge Drive	1015	90210
449302	Pickfair Way	1143	90210
475610	Robert Lane	1004	90210
502070	Waring Avenue	445	90210
550493	Wallace Ridge	1300	90210
556677	Angelo Drive	1280	90210
662288	Franklin Canyon Drive	1900	90210
663322	Chalette Drive	510	90210
667382	Deep Canyon Drive	1805	90210
701030	Spalding Drive	222	90210
774859	Arkell Drive	1401	90210
775599	Hidden Valley Road	2505	90210
801060	Gregory Way	315	90210

DeliveryAddressID	DeliveryAddressStreet	DeliveryAddressStreetNum	DeliveryAddressZip
847561	Doheny Road	501	90210
884756	Summit Drive	1100	90210
902040	**Charleville Boulevard**	**9420**	90210
904857	Heather Road	955	90210
928374	Schuyler Road	912	90210

DeliveryServiceID	DeliveryServiceName	DeliveryServicePhone	DeliveryAddressIDCode
103366	Great Lakes Dispatch	31351818	123847
108822	Silver State Express	70254141	123847
119933	Interstate Expeditors	61255353	123847
120394	Gold Star Trucking	51055531	123847
192038	Blue Dart Deliveries	81853344	123847
206655	Liberty Bell Logistics	21559696	123847
219485	SwiftStream Logistics	31051029	123847
225544	Northwest Cargo Link	50351919	123847
228374	Point-B Logistics	61955509	123847
301144	Peak Performance Mail	25355252	123847
331100	IronClad Shipping	21255575	123847
337711	Canyon View Couriers	80152929	123847
339485	City-to-City Haulage	20651212	123847
447766	Bayou Rapid Transit	50455030	123847
476102	90210 Express Mail	31055510	123847
509977	Sunshine State Deliver	30556363	123847
550192	HyperLink Couriers	32355582	123847
552288	Windy City Couriers	31255686	123847
661029	Pacific West Cargo	80555561	123847
664477	Southern Cross Freight	40550808	123847
708844	Commonwealth Cargo	80455507	123847
773399	Lone Star Logistics	51255797	123847
774821	Velox Parcel Services	21359930	123847
775533	RapidResponse Dispatch	40552020	123847
802266	Alpine Logistics Group	30557474	123847

DeliveryServiceID	DeliveryServiceName	DeliveryServicePhone	DeliveryAddressIDCode
837261	DirectRoute Freight	41555721	123847
884712	MileHigh Courier Co	91655242	123847
904433	Midwest Haulage Co	31455585	123847
991122	NextDay Global Trans	71454433	123847
998811	Red Zone Transport	60255564	123847

ItemID	ItemName	ItemSupplierID	ItemCategory	ItemLeft	ItemCost
102948	Smart Espresso Machine	112233	Appliances	45	1200
119283	Aromatic Soy Candle	112233	Home Decor	850	45
119934	Air Purifying System	112233	Appliances	65	450
120394	Bluetooth Surround Bar	112233	Electronics	115	800
120395	Hand-Woven Area Rug	112233	Home Decor	9	4200
219384	Professional Chef Knife	112233	Kitchenware	156	125
225544	Wine Cellar Cooling Unit	112233	Appliances	7	3200
228374	Stainless Steel Air Fryer	112233	Appliances	128	220
331100	Adjustable Standing Desk	112233	Furniture	27	1350
334102	Crystal Stemware Set	112233	Kitchenware	89	250
339485	Dimmable Floor Lamp	112233	Lighting	43	320
440293	Smart Home Hub	112233	Electronics	412	199
482716	Goose Down Duvet	112233	Bedding	22	600
550192	Cashmere Throw Blanket	112233	Home Decor	67	180
552288	Cordless Stick Vacuum	112233	Appliances	82	700
556611	Organic Cotton Towels	112233	Bath	540	75
592831	Silk Egyptian Sheets	112233	Bedding	142	450
661029	Memory Foam Mattress	112233	Bedding	14	2500
664477	Crystal Chandelier	112233	Lighting	3	4950
667102	Velvet Occasional Chair	112233	Furniture	8	890
771029	Noise-Cancelling Headphones	112233	Electronics	230	350
773322	Ergonomic Office Chair	112233	Furniture	56	1100
773399	Smart Garden Planter	112233	Garden	150	160
775533	Smart Door Lock	112233	Electronics	204	299
884712	Cast Iron Dutch Oven	112233	Kitchenware	94	280

ItemID	ItemName	ItemSupplierID	ItemCategory	ItemLeft	ItemCost
884713	Copper Cookware Set	112233	Kitchenware	18	1500
884729	Teak Outdoor Lounge	112233	Furniture	12	3400
903847	Electric Stand Mixer	112233	Appliances	34	550
991122	Porcelain Dinnerware Set	112233	Kitchenware	31	450
998811	Abstract Oil Painting	112233	Home Decor	5	4800

ReceiptSaleID	ReceiptItemID	ReceiptItemCount
665544	592831	2
665544	102948	1
665544	884729	4
665544	334102	5
665544	771029	2
665544	550192	3
665544	219384	1
665544	667102	1
665544	440293	5
665544	119283	4
665544	903847	1
665544	482716	2
665544	773322	3
665544	120394	1
665544	884712	5
665544	556611	4
665544	339485	2
665544	991122	3
665544	228374	1
665544	661029	2
665544	775533	5
665544	120395	1
665544	884713	2
665544	119934	1
665544	998811	1

ReceiptSaleID	ReceiptItemID	ReceiptItemCount
665544	331100	3
665544	552288	1
665544	773399	4
665544	664477	1
665544	225544	1

SaleID	SaleCustomerID	SaleDeliveryServiceID
101020	108823	331100
105090	108823	331100
112233	108823	331100
123456	108823	331100
123546	108823	331100
192837	108823	331100
204060	108823	331100
221133	108823	331100
223344	108823	331100
302010	108823	331100
332211	108823	331100
334455	108823	331100
339922	108823	331100
415263	108823	331100
446688	108823	331100
475869	108823	331100
506070	108823	331100
554433	108823	331100
654321	108823	331100
665544	108823	331100
667788	108823	331100
705030	108823	331100
774411	108823	331100
778899	108823	331100
807060	108823	331100

SaleID	SaleCustomerID	SaleDeliveryServiceID
889900	108823	331100
905040	108823	331100
987654	108823	331100
990011	108823	331100
998877	108823	331100

SupplierID	SupplierName	SupplierPhone	SupplierAddressIDCode
101020	Silver State Mining	75557744	556201
102938	Golden Gate Wholesale	21358821	556201
105090	Great Lakes Glass	35555050	556201
112233	North Star Industrial	61257788	556201
123546	Global Reach Freight	51052233	556201
192837	Blue Horizon Textiles	81852345	556201
204060	Liberty Bell Fabrics	55553030	556201
221133	Precision Parts Inc	61955344	556201
223344	Evergreen Forestry	55555566	556201
302010	Cascade Equipment	25558833	556201
332211	Empire State Sourcing	21251100	556201
334455	Emerald City Trading	20655122	556201
339922	Canyon Rim Partners	15556060	556201
446688	Bayou Beverage Group	55556655	556201
475869	Beverly Hills Tech Corp	31557766	556201
482910	Apex Logistics Solutions	31550192	556201
506070	Sunshine State Citrus	55559922	556201
554433	Windy City Hardware	31252299	556201
556473	Summit Manufacturing	32355732	556201
665544	Coastal Supply Co	80555577	556201
667788	Peach State Produce	45554477	556201
705030	Old Dominion Paper	45554040	556201
774411	NextGen Electronics	40850011	556201
778899	Lone Star Steel	51255388	556201
807060	Rocky Mountain Gear	35551010	556201

SupplierID	SupplierName	SupplierPhone	SupplierAddressIDCode
827364	Pacific Rim Imports	41555012	556201
889900	Mainland Materials	91655455	556201
905040	Gateway City Freight	45552020	556201
990011	Desert Springs Water	60255500	556201
998877	Starlight Distribution	71558899	556201

SupplierAddressID	SupplierAddressStreet	SupplierAddressStreetNum	SupplierAddressZip
102030	Linden Drive	505	90210
129485	Cielo Drive	10050	90210
192830	Deep Canyon Drive	2112	90210
202134	San Ysidro Drive	1701	90210
204080	Peck Drive	335	90210
221199	Sierra Drive	802	90210
304928	Bowmont Drive	2555	90210
314256	Carla Lane	1018	90210
338271	Benedict Canyon Drive	2615	90210
405060	Roxbury Drive	915	90210
440011	Heather Road	9210	90210
443322	Doheny Road	412	90210
449281	Shadow Hill Way	1225	90210
495817	Lania Lane	1288	90210
510293	Wallace Ridge	1350	90210
556201	Kimridge Road	1390	90210
559182	Loma Vista Drive	1195	90210
664433	Palm Drive	615	90210
665102	Ambassador Avenue	1001	90210
667123	Hidden Valley Road	2480	90210
706050	Canon Drive	422	90210
772109	Hutton Drive	1905	90210
774123	Coldwater Canyon Drive	1845	90210
779900	Maple Drive	710	90210
809010	Bedford Drive	601	90210

SupplierAddressID	SupplierAddressStreet	SupplierAddressStreetNum	SupplierAddressZip
827319	Bundy Drive	1540	90210
883746	Dona Marina Drive	3105	90210
884102	Beverly Estate Drive	1201	90210
901020	Beverly Drive	210	90210
993847	Tower Grove Drive	1140	90210

ZipCode	Country	Region	City
02108	United States	Massachusetts	Boston
0800	Australia	Northern Territory	Darwin
10001	United States	New York	New York
19102	United States	Pennsylvania	Philadelphia
2000	Australia	New South Wales	Sydney
20001	United States	District of Columbia	Washington
20121	Italy	Lombardy	Milan
2600	Australia	ACT	Canberra
3000	Australia	Victoria	Melbourne
30301	United States	Georgia	Atlanta
33101	United States	Florida	Miami
37201	United States	Tennessee	Nashville
4000	Australia	Queensland	Brisbane
43201	United States	Ohio	Columbus
46201	United States	Indiana	Indianapolis
48201	United States	Michigan	Detroit
5000	Australia	South Australia	Adelaide
55401	United States	Minnesota	Minneapolis
6000	Australia	Western Australia	Perth
60601	United States	Illinois	Chicago
64101	United States	Missouri	Kansas City
7000	Australia	Tasmania	Hobart
70001	United States	Louisiana	Metairie
75201	United States	Texas	Dallas
80202	United States	Colorado	Denver

ZipCode	Country	Region	City
85001	United States	Arizona	Phoenix
89101	United States	Nevada	Las Vegas
90210	United States	California	Beverly Hi
94102	United States	California	San Franci
97201	United States	Oregon	Portland
98101	United States	Washington	Seattle

Part 2: Data Manipulation

Requirement f:

```
SELECT CustomerAddress.CustomerAddressID, CustomerAddress.CustomerAddressStreet, CustomerAdd
FROM Customer
INNER JOIN CustomerAddress ON CustomerAddress.CustomerAddressID = Customer.CustomerAddressID
INNER JOIN Zip ON Zip.ZipCode = CustomerAddress.CustomerAddressZip;
```

CustomerAddressID	CustomerAddressStreet	CustomerAddressStreetNum	CustomerAddressZip	CustomerFirstName	CustomerSecondName	CustomerLastName	CustomerPhone	Country	Region	City	CustomerAddressIDCode
582914	Via Dante	14	20121	Bruno	Alfonso	Farina	302555326	Italy	Lombardy	Milan	582914
582914	Via Dante	14	20121	Antonio	Pasquale	Esposito	390555302	Italy	Lombardy	Milan	582914
582914	Via Dante	14	20121	Flavio	Valerio	Rinaldi	390255319	Italy	Lombardy	Milan	582914
582914	Via Dante	14	20121	Carmine	Renato	Costa	390255311	Italy	Lombardy	Milan	582914
582914	Via Dante	14	20121	Fabrizio	Paolo	De Luca	390255309	Italy	Lombardy	Milan	582914
582914	Via Dante	14	20121	Renzo	Ivano	Pellegrini	302555324	Italy	Lombardy	Milan	582914
582914	Via Dante	14	20121	Augusto	Tullio	Santoro	392555317	Italy	Lombardy	Milan	582914
582914	Via Dante	14	20121	Giuseppe	Franco	Bruno	390255306	Italy	Lombardy	Milan	582914
582914	Via Dante	14	20121	Giacomo	Ettore	Caruso	390255320	Italy	Lombardy	Milan	582914
582914	Via Dante	14	20121	Giorgio	Filippo	Rizzo	390255513	Italy	Lombardy	Milan	582914
582914	Via Dante	14	20121	Carlo	Arturo	Ferrero	902555327	Italy	Lombardy	Milan	582914
582914	Via Dante	14	20121	Salvatore	Angelo	Marino	390255304	Italy	Lombardy	Milan	582914
582914	Via Dante	14	20121	Dario	Duccio	Serra	390255328	Italy	Lombardy	Milan	582914
582914	Via Dante	14	20121	Mauro	Silvio	Mariani	390555318	Italy	Lombardy	Milan	582914
582914	Via Dante	14	20121	Luciano	Nico	Ferraro	390255321	Italy	Lombardy	Milan	582914
582914	Via Dante	14	20121	Raffaele	Umberto	Lombardi	392555314	Italy	Lombardy	Milan	582914
582914	Via Dante	14	20121	Vincenzo	Domenico	Ricci	390255303	Italy	Lombardy	Milan	582914
582914	Via Dante	14	20121	Fausto	Lio	Vitale	390255329	Italy	Lombardy	Milan	582914
582914	Via Dante	14	20121	Claudio	Massimo	Gallo	390255307	Italy	Lombardy	Milan	582914
582914	Via Dante	14	20121	Igor	Zeno	Messina	390555330	Italy	Lombardy	Milan	582914
582914	Via Dante	14	20121	Emilio	Cesare	Fontana	390255316	Italy	Lombardy	Milan	582914
582914	Via Dante	14	20121	Aldo	Nino	Sanna	390255525	Italy	Lombardy	Milan	582914
582914	Via Dante	14	20121	Stefano	Roberto	Greco	390255305	Italy	Lombardy	Milan	582914
582914	Via Dante	14	20121	Pietro	Rosario	Barbieri	390555315	Italy	Lombardy	Milan	582914
582914	Via Dante	14	20121	Daniele	Enrico	Conti	390255508	Italy	Lombardy	Milan	582914
582914	Via Dante	14	20121	Barry	The	Bot	720137410	Italy	Lombardy	Milan	582914
582914	Via Dante	14	20121	Enzo	Ottavio	Valli	390255522	Italy	Lombardy	Milan	582914
582914	Via Dante	14	20121	Giovanni	Battista	Moretti	390555301	Italy	Lombardy	Milan	582914
582914	Via Dante	14	20121	Gianluca	Sandro	Mancini	390255310	Italy	Lombardy	Milan	582914
582914	Via Dante	14	20121	Paolo	Santi	Guerra	390255323	Italy	Lombardy	Milan	582914
582914	Via Dante	14	20121	Giovanni	Battista	Moretti	392555231	Italy	Lombardy	Milan	582914
582914	Via Dante	14	20121	Pasquale	Gino	Giordano	39025512	Italy	Lombardy	Milan	582914

Figure 14: img

Requirement g:

```
SELECT DISTINCT Country
FROM Zip;
```

Requirement h:

```
SELECT CustomerFirstName, CustomerSecondName, CustomerLastName, CustomerID
FROM Customer
WHERE CustomerLastName LIKE '%ot%'
```

Requirement i:

```
SELECT CustomerFirstName, CustomerSecondName, CustomerLastName, Sale.SaleID, Zip.Country, Sa
```

Country
United States
Australia
Italy

Figure 15: img

CustomerFirstName	CustomerSecondName	CustomerLastName	CustomerID
Barry	The	Bot	789234

Figure 16: img

```
FROM Customer
INNER JOIN CustomerAddress ON CustomerAddress.CustomerAddressID = Customer.CustomerAddressID
INNER JOIN Zip ON Zip.ZipCode = CustomerAddress.CustomerAddressZip
INNER JOIN Sale ON Sale.SaleCustomerID = Customer.CustomerID
WHERE Zip.Country LIKE 'Italy'
```

Requirement j:

```
SELECT ItemID, ItemName, ItemCost,
        CASE
                WHEN ItemCost > 100 AND ItemCost < 200 THEN ItemCost*0.9
        END AS ItemCostAfterDiscount
FROM Item
WHERE ItemCost > 100 AND ItemCost < 200
```

AI Decleration

I have used AI while undertaking my assignment in the following ways:

- To develop research questions on the topic – YES/ **NO**
- To create an outline of the topic – YES/ **NO**
- To explain concepts – YES/ **NO**
- To support my use of language – YES/ **NO**

CustomerFirstName	CustomerSecondName	CustomerLastName	SaleID	Country	SaleCustomerID	CustomerAddressID	CustomerAddressZip	ZipCode	SaleCustomerID	CustomerAddressIDCode
Flavio	Valerio	Rinaldi	101020	Italy	108823	582914	20121	20121	108823	582914
Flavio	Valerio	Rinaldi	105090	Italy	108823	582914	20121	20121	108823	582914
Flavio	Valerio	Rinaldi	112233	Italy	108823	582914	20121	20121	108823	582914
Flavio	Valerio	Rinaldi	123456	Italy	108823	582914	20121	20121	108823	582914
Flavio	Valerio	Rinaldi	123546	Italy	108823	582914	20121	20121	108823	582914
Flavio	Valerio	Rinaldi	192837	Italy	108823	582914	20121	20121	108823	582914
Flavio	Valerio	Rinaldi	204060	Italy	108823	582914	20121	20121	108823	582914
Flavio	Valerio	Rinaldi	221133	Italy	108823	582914	20121	20121	108823	582914
Flavio	Valerio	Rinaldi	223344	Italy	108823	582914	20121	20121	108823	582914
Flavio	Valerio	Rinaldi	302010	Italy	108823	582914	20121	20121	108823	582914
Flavio	Valerio	Rinaldi	332211	Italy	108823	582914	20121	20121	108823	582914
Flavio	Valerio	Rinaldi	334455	Italy	108823	582914	20121	20121	108823	582914
Flavio	Valerio	Rinaldi	339922	Italy	108823	582914	20121	20121	108823	582914
Flavio	Valerio	Rinaldi	415263	Italy	108823	582914	20121	20121	108823	582914
Flavio	Valerio	Rinaldi	446688	Italy	108823	582914	20121	20121	108823	582914
Flavio	Valerio	Rinaldi	475869	Italy	108823	582914	20121	20121	108823	582914
Flavio	Valerio	Rinaldi	506070	Italy	108823	582914	20121	20121	108823	582914
Flavio	Valerio	Rinaldi	554433	Italy	108823	582914	20121	20121	108823	582914
Flavio	Valerio	Rinaldi	654321	Italy	108823	582914	20121	20121	108823	582914
Flavio	Valerio	Rinaldi	665544	Italy	108823	582914	20121	20121	108823	582914
Flavio	Valerio	Rinaldi	667788	Italy	108823	582914	20121	20121	108823	582914
Flavio	Valerio	Rinaldi	705030	Italy	108823	582914	20121	20121	108823	582914
Flavio	Valerio	Rinaldi	774411	Italy	108823	582914	20121	20121	108823	582914
Flavio	Valerio	Rinaldi	778899	Italy	108823	582914	20121	20121	108823	582914
Flavio	Valerio	Rinaldi	807060	Italy	108823	582914	20121	20121	108823	582914
Flavio	Valerio	Rinaldi	889900	Italy	108823	582914	20121	20121	108823	582914
Flavio	Valerio	Rinaldi	905040	Italy	108823	582914	20121	20121	108823	582914
Flavio	Valerio	Rinaldi	987654	Italy	108823	582914	20121	20121	108823	582914
Flavio	Valerio	Rinaldi	990011	Italy	108823	582914	20121	20121	108823	582914
Flavio	Valerio	Rinaldi	998877	Italy	108823	582914	20121	20121	108823	582914

Figure 17: img

ItemID	ItemName	ItemCost	ItemCostAfterDiscount
219384	Professional Chef Knife	125	112.5
440293	Smart Home Hub	199	179.1
550192	Cashmere Throw Blanket	180	162.0
773399	Smart Garden Planter	160	144.0

Figure 18: img

- To summarise the following articles/resources: **NONE**
- In other ways, as described below:
  - To create dummy data for me to fill into tables for the Part 1's 5th question, using Gemini AI.